

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 4
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GitHub: <https://github.com/Prabh-84/streaming-rag>

Theme

Streaming Live RAG for Real-Time

Challenge: Traditional RAG waits for a complete query before retrieval.

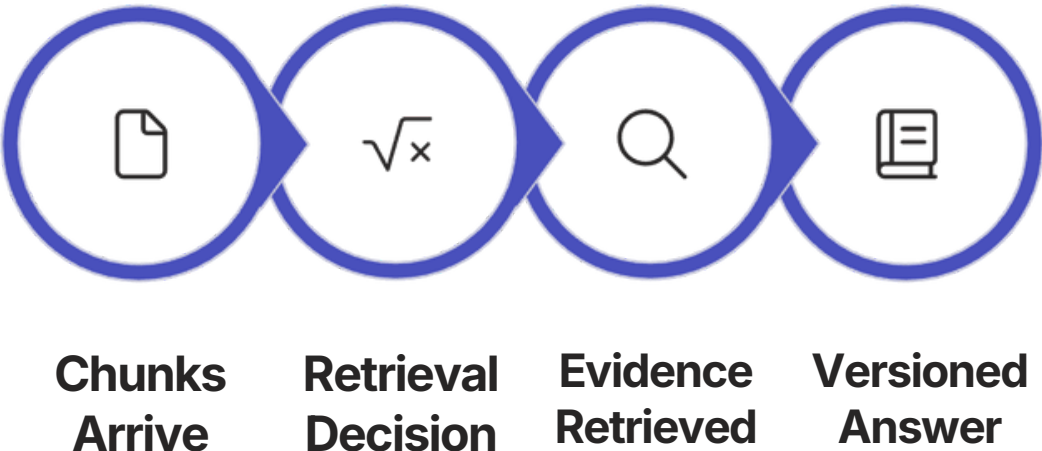
- **Our approach:** Retrieve relevant evidence while the user's transcript is still streaming.
- **Key capabilities:**
 - Early retrieval from partial context
 - Multi-intent query decomposition
 - Parallel dense + sparse retrieval
 - Evidence fusion and reranking
 - Session-aware answer refinement
 - Grounded answers with citations

Existing Solutions & Gaps

Conventional batch RAG waits for a complete utterance before retrieving anything. Streaming Live RAG makes retrieval decisions from partial input, decomposing compound questions and incorporating corrections without restarting from scratch.



Streaming Live RAG



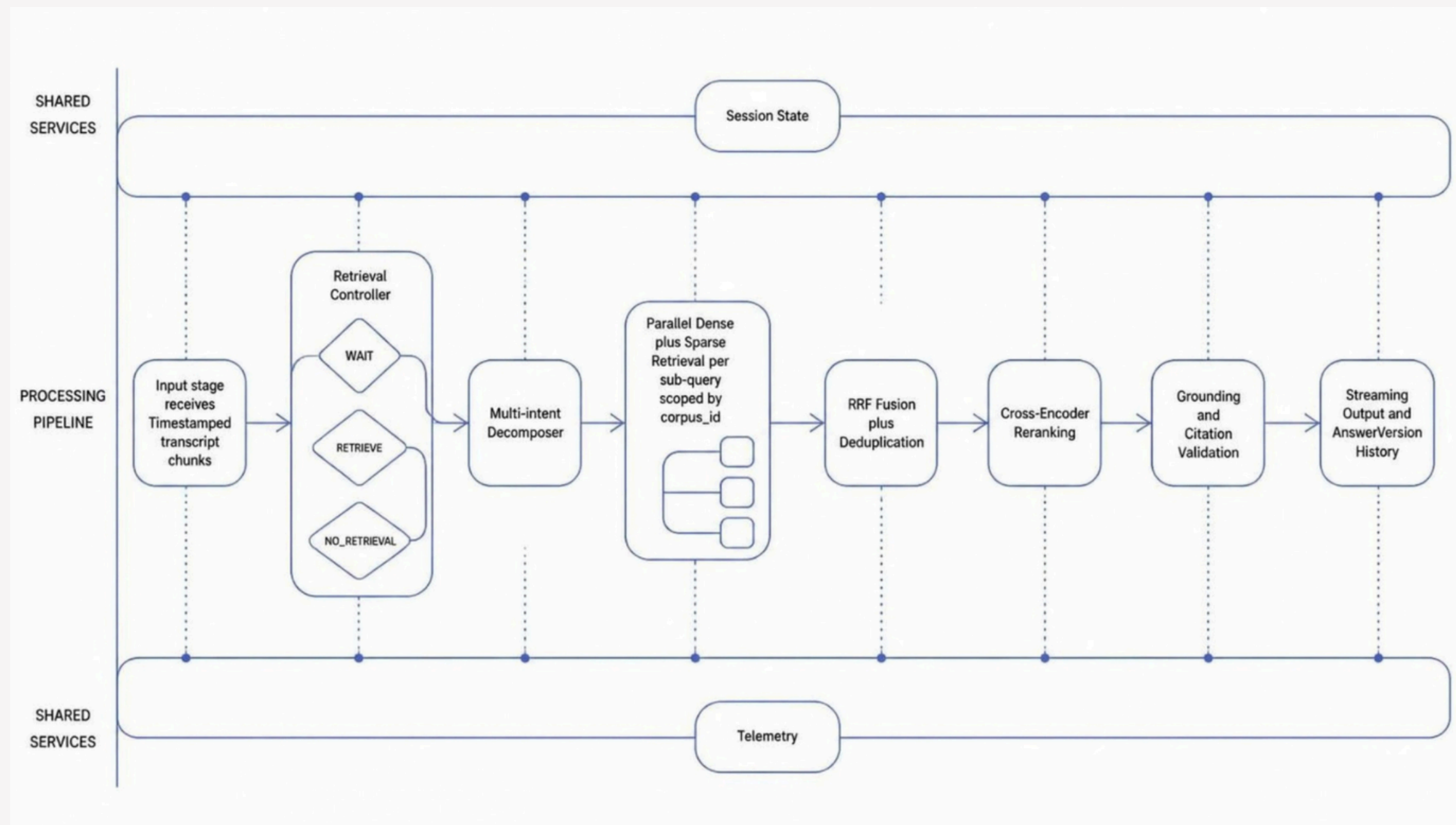
Gaps Scoped to a Conventional Batch Baseline

Dimension	Batch Baseline	Streaming Live RAG
Timing	Waits for a complete question	Decides from partial input
Decomposition	One broad query	Independently searchable sub-queries
Refinement	Repeats the full search	Updates only relevant work
Grounding	Opaque answers	Traceable citations and explicit uncertainty



Our Solutions & Architecture Diagram

One FastAPI service coordinates all components through rule-based orchestration and scoped LLM calls. Rule-based decisions govern retrieval timing; LLM calls handle structured decomposition and answer generation.



Rule-Based Components

- Retrieval controller: WAIT / RETRIEVE / NO_RETRIEVAL decisions
- RRF fusion and deduplication
- Grounding and citation validation
- Session state management

LLM-Assisted Components

- Multi-intent decomposition into sub-queries
- Answer generation with structured output
- Uncertainty fallback when evidence is insufficient
- Structured telemetry logging

Demo & Product Walkthrough

northstar_demo_extended Corpus

The demo uses the **northstar_demo_extended** corpus. The four moments below trace a single compound query from streaming input to versioned, cited output.

1

Stream Transcript Chunks

Timestamped transcript chunks arrive incrementally. The retrieval controller evaluates each chunk and decides whether to WAIT, RETRIEVE, or apply NO_RETRIEVAL.

2

Detect Two Intents, Four Retrieval Calls

The multi-intent decomposer identifies two sub-queries. Each sub-query triggers one dense retrieval call and one sparse retrieval call, totaling four parallel calls scoped by corpus_id.

3

Differentiated Outcomes

The hostel visitor-policy question returns a grounded, cited answer. The library-fine question returns explicit uncertainty because retrieved evidence does not support a specific fine claim.

4

Inspect Evidence and Telemetry

The evidence panel, Answer Version 1 card, citation mapping, and telemetry dashboard are all visible and inspectable in the browser frontend.

 Authentic screenshots should replace these placeholders before submission. Never fabricate screenshots or invent evidence panel content.

Tools and tech stack used



Interface

- React 18
- TypeScript
- Vite
- CSS Modules



Language Model

- Groq serving openai/gpt-oss-120b
- Not an OpenAI API integration



Reranking and NLP

- cross-encoder/ms-marco-MiniLM-L-6-v2
- spaCy en_core_web_sm



API and Validation

- FastAPI
- Pydantic v2
- Python 3.11+



Retrieval

- Qdrant (dense vector store)
- BM25 (sparse retrieval)
- BAAI/bge-small-en-v1.5 embeddings
- RRF fusion



Deployment and Quality

- Docker and Docker Compose
- pytest and pytest-asyncio
- ruff (linting)

Impact & Use case

Consider a support caller who adds another question or corrects a constraint while still speaking. Streaming Live RAG handles this without restarting the retrieval pipeline from scratch.

Expected Benefits

Lower Perceived Latency

Retrieval can start before the question finishes, reducing the time a caller waits for a first response.

Correctly Scoped Searches

Compound questions become independent, correctly scoped searches rather than one broad query that may miss intent.

Selective Refinement

Corrections update only the relevant work instead of restarting everything, preserving prior evidence and answer history.

Traceable Factual Sentences

Every factual sentence in the answer remains traceable to corpus evidence through explicit citation mapping.

Potential Applications



These are potential applications. No customer deployments are claimed.

Live Voice Support

Real-time agent assistance during inbound calls

Agent Copilot

Parallel retrieval while the agent listens

Campus / HR / Compliance

Policy lookup from structured internal corpora

Internal Knowledge Base

Innovation highlights, results and limitations

Engineering Contributions



Early Retrieval

Retrieval on partial transcript before utterance completes



Delta Refinement

Session-aware updates preserving prior AnswerVersion history

Reported Results

411

Tests Passed

3 skipped, 0 failed in the test suite

10/10

Offline Scenarios

All benchmark scenarios passed

- Real Groq end-to-end run verified
- Real browser frontend end-to-end run verified
- Corpus isolation, citation/evidence mapping, and Docker deployment verified in the supplied project report



Multi-Intent Decomposition

Deterministic controller plus structured LLM decomposition



Strict Grounding

Uncertainty fallback and structured replayable telemetry

Limitations

Citation-Support Measurement

The current harness does not automatically measure the 85% citation-support percentage target.

Fabrication Rate Caveat

Measured fabrication rate was 0% in evaluated cases. This result must not be generalized to all inputs.

Cold-Start Latency

Latency can increase while embedding and reranker models download or warm up.

Simulated Streaming

Transcript input is simulated streaming text, not production ASR.

What's next

Roadmap — Future Work

- **Production ASR Integration**
Replace simulated streaming text with a live automatic speech recognition feed.
- **Broader Multi-Domain Corpora**
Extend corpus coverage beyond the current demo domain.
- **Persistent Model Caching**
Expanded telemetry and monitoring for production traffic volumes.
- **Stronger Automated Grounding Metrics**
Automated measurement of citation-support percentage at scale.
- **Production-Scale Observability**
Optimized serving to eliminate cold-start latency from model downloads.
- **Runtime Corpus Management**
Optional workflow for adding or updating corpora without redeployment.

Differentiation — Why Each Group Matters

Early Retrieval and Compound-Intent Handling

Reduces perceived latency and ensures each intent is searched independently, improving recall for multi-part questions.

Selective Refinement with AnswerVersion History

Corrections update only the affected sub-query, preserving prior evidence and enabling users to compare answer versions.

Retrieval-Layer Corpus Isolation and Citation Validation

Each sub-query is scoped to a specific corpus_id, preventing cross-corpus contamination and ensuring every factual claim is traceable.

Deterministic Identifiers, Telemetry, and Reproducible Deployment

Structured telemetry and Docker Compose deployment make every run reproducible and auditable without manual intervention.

Checklist — Updated on Public GitHub

All items below reflect the state of the supplied project materials. Items marked with a placeholder require team action before final submission.

Deliverable	Status	Reference
Working prototype code	YES — per supplied project materials	Commit 8e15d5b
README with reproducible setup instructions	YES — per supplied project materials	Commit 8e15d5b
Demo video (max 5 minutes)	https://drive.google.com/file/d/1YIH1CKF1u0_f08KOWih6ia3dZTISk0R2/view?usp=sharing	Team action required
Presentation file	YES	Team action required
Public GitHub repository	YES	github.com/Prabh-84/streaming-rag
Reference commit recorded	YES — per supplied project materials	8e15d5b

THANK YOU